Dhafer Malouche, Ph.D.,
Professor at the University of Carthage

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https://dhafermalouche.net/ https://github.com/malouche

Google scholar page

Employment History

05/2020 – 01/2021

World Health Organization, Tunisia Senior data analyst consultant, Measuring the impact of the COVID 19 pandemic on psychiatric needs of the general population in Tunisia, data collected from a hot line accessible throughout the country, including those without access to Internet.

WARC, Africa, Sierra Leon Data analyst consultant, Setting up an online daily data monitoring platform for a survey implemented in Sierra Lean.

2003 – · · · ·

University of Carthage, Ecole Supérieure de la Statistique et de l'Analyse de l'Information, Tunis, Tunisia

- **Positions:** Professor (from 2018), Associate Professor (2011-2018), Assistant Professor (2003-2011).
- **Teaching:** Data Analysis: Principal Component Analysis, Correspondence Analysis, and Multiple Correspondence Analysis, Theory and Practice with `R`, Mathematical Statistics: Statistical Inference, Hypothesis Testing, Regression Analysis, Theory and Practice with `R`, Data Mining and Practice with `R`, Bayesian Statistics with `OpenBugs/Winbugs/RStan/Jags`, Time Series: ARMA and SARIMA Processes, Theory and Practice with `R/Python`, Big Data: Large Data with `R/Python`, `SQL`, `Spark`, `H2O`, Advanced `R/Python`: Data Management, Data Visualization, Dashboards, Shiny Apps, Heroku Apps, Bokeh.
- **Research:**
  - Supervising 5 Ph.D.: Detection and classification of swallowing sound, Sensory Analysis, Genetics, Zoonotic cutaneous leishmaniasis incidence, Cardiovascular risk factor.
  - Papers and collaborations: I had published and coauthored more than 40 scholar papers.
  - Supervising Master and Engineering Thesis on several topics related to applied statistics and data science.
- **Administration:** Director of the Department of Statistics (2004-2007)

2014 – 2019

Yale University, Whitney and Betty MacMillan Center for International and Area Studies and the Department of Statistics and Data Science, New Haven, USA.

- **Positions:** Visiting Associate Professor (2014), Consulting on several projects with Yale Scholars (2015-2018), Associate Research Scholar (2019).
- **Missions:** Democratic Transition in Tunisia, Governance and Local Development: Implementing two face-o-face Surveys in Tunisia, G-econ team on building local GDP data,Teaching Time Series with `R/Python Course`.

Employment History (continued)

2016 – 2017

University of Michigan, Center for Political Studies of the Institute for Social Research, Ann Arbor, USA.

- **Position:** Visiting Fulbright Scholar.
- **Role:** Research: Working on Data Quality, Survey Methodology, Interviewer Effect, Teaching: 4 Lectures on Applied graphical Models, 1 Lecture on Data visualization with `R`, 1 Lecture on Sensory Analysis.

2011 (May to July)

Stanford University, Department of Statistics, Palo alto, USA.

- **Position:** Visiting Fulbright Scholar.
- **Role:** Research on Graphical Models, Faithfulness assumption, and Covariance graphs.

2002 – 2003

York University, Department of Statistics, Toronto, Canada.

- **Position:** Visiting Assistant Professor,
- **Teaching:** Applied Regression Models with SAS, Introduction to the theory of probability, Introduction to Statistics with Minitab,
- **Research:** Monte Carlo Methods and Bayesian Estimation of the Graphical Models.

1998 – 2002

University of Sousse , Institut Préparatoire aux Ecoles d'Ingénieurs, Sousse, Tunis.

- Teaching: Analysis, Calculus, Algebra
- Research: Natural Exponential Families, Pick functions, Markov Chains
- Administration: Director of the Department of Mathematics

Education

Sept 2009

Habilitation *(Tenure)*, Université de Tunis El Manar Ecole National d'Ingénieurs de Tunis, Tunisia.
Thesis title: *Problèmes autour de la probabilité et de la statistique: Méthodes et Applications.*
Dissertation: `https://malouche.github.io/myCV/reports.html`

October 1997

Doctorate, (Ph.D.), Paul Sabatier University, Toulouse, France.
Thesis title: *Classification des familles exponentielles associées à des fonctions Pick.*
Dissertation: `https://malouche.github.io/myCV/reports.html`

1993–1994

Master's Degree (D.E.A), Paul Sabatier University, Applied Mathematics.

1989 – 1993

Bachelor (Maîtrise), Ecole Normale Supérieure de Bizerte, Tunisia.

Skills

Languages	Strong reading, writing and speaking competencies for English, French, and Arabic.
Coding	`Python`, `R`, TABLEAU, SPARK, H2O, `Shiny`, LaTeX, …
Misc.	Data Mining, Machine Learning, Big Data, Academic research, Teaching, training, consultation, LaTeX typesetting and publishing.

Research Publications (Most recent papers)

1. Lahmandi-Ayed, R., & Malouche, D. (2020). More investment in research and development for better education in the future? *Available at SSRN 3739632*.
2. Malouche, D., & Ben Romdhane, H. (2020). Estimating excess mortality associated with covid-19 pandemic: A 151 cross-countries study. *Estimating Excess Mortality Associated with COVID-19 Pandemic: A*, *151*.
3. Saidi, O., Malouche, D., Saksena, P., Arfaoui, L., Talmoudi, K., Hchaichi, A., … Ben Alaya, N. (2020). Impact of contact tracing, respect of isolation and lockdown in reducing the number of cases infected with covid-19: Case study: Tunisia's response from march 22 to 04 may 2020. *International Journal of Infectious Diseases*.
4. Kongbonga, G. Y. M., Hassine, K. B., Ghalila, H., Malouche, D. et al. (2019). Front-face fluorescence using uv-led coupled to usb spectrometer to discriminate between virgin olive oil from two cultivars. *Food and Nutrition Sciences*, *10*(02), 119.
5. Saidi, O., Hajjem, S., Zoghlami, N., Aounallah-Skhiri, H., Mansour, N. B., Hsairi, M., … O'Flaherty, M. et al. (2019). Premature mortality attributable to smoking among tunisian men in 2009. *Tobacco induced diseases*, *17*.
6. Saidi, O., O'Flaherty, M., Zoghlami, N., Malouche, D., Capewell, S., Critchley, J. A., … Guzman Castillo, M. (2019). Comparing strategies to prevent stroke and ischemic heart disease in the tunisian population: Markov modeling approach using a comprehensive sensitivity analysis algorithm. *Computational and mathematical methods in medicine*, *2019*.
7. Saidi, O., Zoghlami, N., Bennett, K. E., Mosquera, P. A., Malouche, D., Capewell, S., … O'Flaherty, M. (2019). Explaining income-related inequalities in cardiovascular risk factors in tunisian adults during the last decade: Comparison of sensitivity analysis of logistic regression and wagstaff decomposition analysis. *International journal for equity in health*, *18*(1), 1–11.
8. Salem, S., Malouche, D., & Romdhane, H. B. (2019). Tunisian population quality of life: A general analysis using sf-36. *Eastern Mediterranean Health Journal*, *25*(9).
9. Benstead, L. J., Kao, K., Landry, P. F., Lust, E. M., & Malouche, D. (2017). Using tablet computers to implement surveys in challenging environments. *Survey Practice*, *10*(2), 1–9.
10. Rebhi, I., & Malouche, D. (2017a). An approach for external preference mapping improvement by denoising consumer rating data. *INTERNATIONAL JOURNAL OF ADVANCED COMPUTER SCIENCE AND APPLICATIONS*, *8*(12), 500–508.
11. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017). Modeling zoonotic cutaneous leishmaniasis incidence in central tunisia from 2009-2015: Forecasting models using climate variables as predictors. *PLoS neglected tropical diseases*, *11*(8), e0005844.
12. Triki, H. Z., Laabir, M., Lafabrie, C., Malouche, D., Bancon-Montigny, C., Gonzalez, C., … Daly-Yahia, O. K. (2017). Do the levels of industrial pollutants influence the distribution and abundance of dinoflagellate cysts in the recently-deposited sediment of a mediterranean coastal ecosystem? *Science of the Total Environment*, *595*, 380–392.